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● *Original Contribution*

FULLY AUTOMATIC PLAQUE SEGMENTATION IN 3-D CAROTID ULTRASOUND IMAGES

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Abstract—Automatic segmentation of the carotid plaques from ultrasound images has been shown to be an important task for monitoring progression and regression of carotid atherosclerosis. Considering the complex structure and heterogeneity of plaques, a fully automatic segmentation method based on media-adventitia and lumen-intima boundary priors is proposed. This method combines image intensity with structure information in both initialization and a level-set evolution process. Algorithm accuracy was examined on the common carotid artery part of 26 3-D carotid ultrasound images (34 plaques ranging in volume from 2.5 to 456 mm^3) by comparing the results of our algorithm with manual segmentations of two experts. Evaluation results indicated that the algorithm yielded total plaque volume (TPV) differences of −5.3 ± 12.7 and −8.5 ± 13.8 mm^3 and absolute TPV differences of 9.9 ± 9.5 and 11.8 ± 11.1 mm^3. Moreover, high correlation coefficients in generating TPV (0.993 and 0.992) between algorithm results and both sets of manual results were obtained. The automatic method provides a reliable way to segment carotid plaque in 3-D ultrasound images and can be used in clinical practice to estimate plaque measurements for management of carotid atherosclerosis. (E-mail: myding@hust.edu.cn) © 2013 World Federation for Ultrasound in Medicine & Biology.

Key Words: **Segmentation, Carotid plaque, 3-D ultrasound, Level set.**

INTRODUCTION

Stroke is the second leading cause of death among both urban and rural residents of China (Ministry of Health 2006). About 1.5 to 2 million new strokes occur each year in China (Liu et al. 2007; Shi et al. 1989). Many strokes associated with carotid atherosclerosis can be prevented by surgical or non-surgical treatment (Gorelick 1994; Spence 2007). Hence, measurement of carotid atherosclerosis burden, progression and regression in response to non-surgical therapy is important for the management of patients at risk for stroke.

Many studies have explored the use of 3-D ultrasound (US) techniques for monitoring carotid atherosclerosis progression and regression (Fenster et al. 2006; Schminke et al. 2000; Serena 1999) and have shown it to be an accurate and reproducible tool. Compared with conventional 2-D US, 3-D US improves the visualization and quantification of carotid plaque volume and morphology and allows the monitoring of progression of atherosclerosis (Delcker and Diener 1994; Delcker and Tegeler 1998; Palombo et al. 1998; Steinke and Hennerici 1989). These capabilities have the potential to improve and accelerate the development of anti-atherosclerotic therapies.

Several algorithms for carotid plaque segmentation from ultrasound images have been proposed. The majority of these used longitudinal carotid images (Delsanto et al. 2007; Destrempes et al. 2011; Loizou et al. 2007; Shankar et al. 2003). Loizou et al. (2007) segmented plaques in longitudinal images using various snake models, with initialization based on Doppler blood flow images. Delsanto et al. (2007) combined gradient-based segmentation, a snake method and a fuzzy K-means algorithm, with an initialization based on pixel intensity to segment plaque-plus-vessel complex in

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longitudinal images. Destrempes et al. (2011) proposed motion estimation and a Bayesian framework to perform segmentation of plaques using radiofrequency information from longitudinal images. Because in carotid US images the contrast is lower at boundaries parallel to the US beam than boundaries orthogonal to the US beam, the applicability of analysis techniques used for 3-D US generated by acquisition of longitudinal images will not be optimal at the lateral sides of the vessel.

Existing literature on plaque segmentation in transverse US images is limited. Buchanan et al. (2012) developed a semi-automated method incorporating three user-selected boundaries in two views of the 3-D US image to generate geometric approximation of total plaque volume (TPV). Seabra et al. (2009) proposed a novel method based on 3-D freehand ultrasound to provide plaque echo morphology information, in which the plaque was semi-automatically segmented by manually initializing its contour in the first image and then using 2-D active contour segmentation for the other images. In their work, the segmentation method was examined for a total of five carotid arteries without comparison to the manual segmentations. The methods described in the aforementioned articles require many operator interactions, which introduce operator variability.

Other articles have described carotid segmentation techniques in transverse US images, but only of the walls and not of the plaque itself (Gill et al. 2000; Mao et al. 2000; Ukwatta et al. 2011; Zahalka and Fenster 2011).

In this report, we introduce a new automatic algorithm for contouring carotid plaques in 3-D US images to monitor carotid atherosclerosis. This algorithm is based on a hybrid level-set framework and makes use of the media-adventitia boundary (MAB) and the lumen-intima boundary (LIB) as priors. Using these priors, the segmentation of plaques is converted to the task of differentiating normal vessel wall and plaque complex based on the fact that plaques are located within or adjacent to the arterial wall. Use of the MAB and LIB segmented priors is justified by their successful segmentations in reported algorithms (Destrempes et al. 2009; Guerrero et al. 2007; Kovalski et al. 2000; Ukwatta et al. 2011). In our validation studies, the algorithm was tested in the common carotid arteries (CCAs), including the bifurcations. In line with Ukwatta et al. (2011), we focus on segmenting plaques in the CCA as a method for monitoring plaque burden in the CCA. Further work will focus on inclusion of the internal carotid artery (ICA).

METHODS

Study patients

Twenty-six 3-D carotid ultrasound images were obtained from 13 patients with carotid stenosis >60% (according to carotid Doppler flow velocities). Patients provided written informed consent to the study protocol, which was approved by the University of Western Ontario Standing Board of Human Research Ethics. All patients were recruited from the Premature Atherosclerosis Clinic and the Stroke Prevention Clinic at University Hospital (London Health Sciences Centre, London, ON, Canada) and the Stroke Prevention and Atherosclerosis Research Centre (Robarts Research Institute, London, ON, Canada).

3-D ultrasound data acquisition

The 3-D US system used in our study has been described in detail elsewhere (Fenster et al. 2001) and is summarized as follows. The 3-D US images were acquired using an ATL HDI 5000 US machine by translating a 50-mm L12-5 transducer (ATL-Philips, Bothell, WA, USA). The transducer was mounted on a linear motor assembly, which was moved along the lateral side of the neck at a constant speed of 3 mm/s for approximately 4.0 cm. In this study, no cardiac gating was used because vessel walls are less compliant for patients with significant atherosclerosis and exhibit few or no radial changes during the cardiac cycle (Mallett et al. 2009; Reneman et al. 1986). The acquired 2-D images were parallel to each other with a pixel size of 0.1×0.1 mm^2 and separated by 0.2 mm. The 3-D US images were reconstructed as the 2-D images were acquired, resulting in an image size of $0.1 \times 0.1 \times 0.2$ mm^3. To obtain 2-D transverse images for segmentation, the 3-D US images were re-sliced at 1-mm intervals in transverse planes that were perpendicular to the longitudinal axis of the CCA.

Reference plaque boundaries

Manual segmentations. Expert manual segmentations were used as a surrogate for reference. A total of 379 transverse slices from 26 3-D carotid images were manually traced by two experienced radiologists. Each was blinded to the automatic segmentation and to each other's results. Each observer segmented each plaque boundary in a 2-D transverse image once in one segmentation session. This procedure was repeated for five sessions over 8 wk to avoid bias from memory of the plaque boundary. The experts reviewed all images, identified those in which the plaques were obvious and segmented them. They did not draw the boundary if the plaque was not present or could not be identified because of excessive noise or shadowing caused by calcifications. If there was disagreement about plaque presence, the experts then were asked to revise their segmentations together to reach a consensus. Of the 379 transverse slices, 262 slices were identified as having plaques (34 plaques in total). Sixty slices had two plaque boundaries. Figure 1 is the

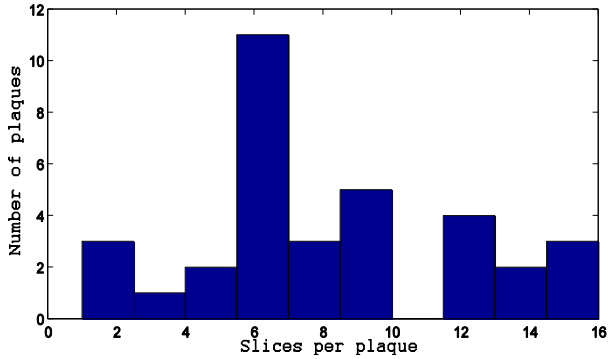

Fig. 1. Histogram of the number of slices per plaque for the 34 plaques.

histogram of the number of slices (*i.e.,* longitudinal length of plaque) for the 34 plaques.

Mean boundary computation. The modified symmetric correspondence algorithm (Chiu et al. 2008a, 2008b) was used to obtain the mean boundary for a single plaque in a transverse image using the five repeated segmentations generated by each observer. This correspondence method could overcome mismatching problems caused by complicated shapes using the original symmetric correspondence algorithm (Papademetris et al. 2001). The validation was performed between algorithm-generated boundaries and computed mean manual boundaries from the two experts, which were taken as references.

Plaque segmentation

Our method makes use of the MAB and LIB contours as prior knowledge for plaque segmentation, which can be obtained either manually or semi-automatically. We used manual segmentations of MAB and LIB, which were then followed by the two steps described below to obtain the plaque boundary. The final segmented contour in the level-set process is the plaque "outer" contour and the region between this "outer" contour and LIB is defined as a plaque region.

1. Initialization. A. Boundary point matching of MAB and LIB. The points on the MAB and LIB contours were matched using the modified symmetric correspondence algorithm proposed by Chiu et al. (2008) to account for irregular shapes of the LIB and cases in which the carotid bifurcation appeared in the 2-D slices. The MAB was divided into about 100 equally spaced points (depending on shape and size), and then the corresponding points on the LIB were identified. This step terminated when all corresponding pairs of points were acquired, as represented by the solid lines in Figure 2c.

B. Initial plaque "outer" contour. Considering that the carotid plaque is more heterogeneous than normal

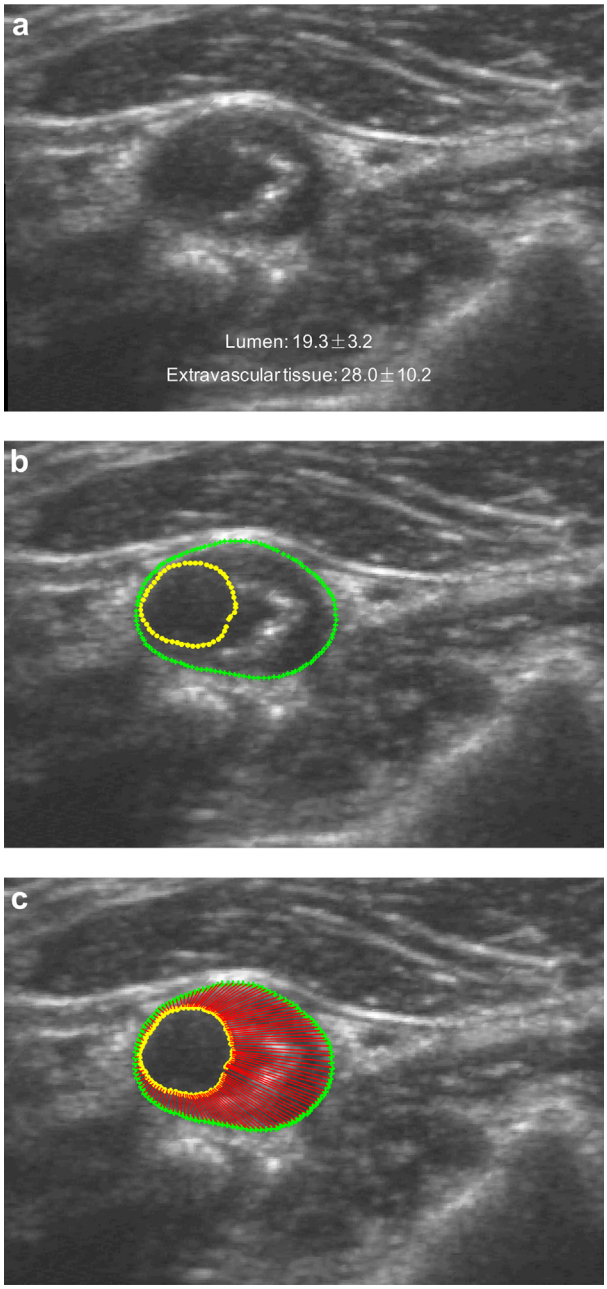

Fig. 2. Example of point correspondence result: (a) original image; (b) image with prior knowledge, that is, lumen-intima boundary in yellow and media-adventitia boundary in green; and (c) point correspondence result, line segments between corresponding points in red.

vessel tissue and is adjacent to the normal tissue, it is more convenient to detect the initial point from MAB to LIB and not in the opposite direction. The initial plaque "outer" points were acquired using the contextual gray-level information between MAB and LIB. In detail, the acquisition was processed using two steps.

Step 1: Polynomial fit to gray levels along correspondence lines. After obtaining all corresponding pairs, the

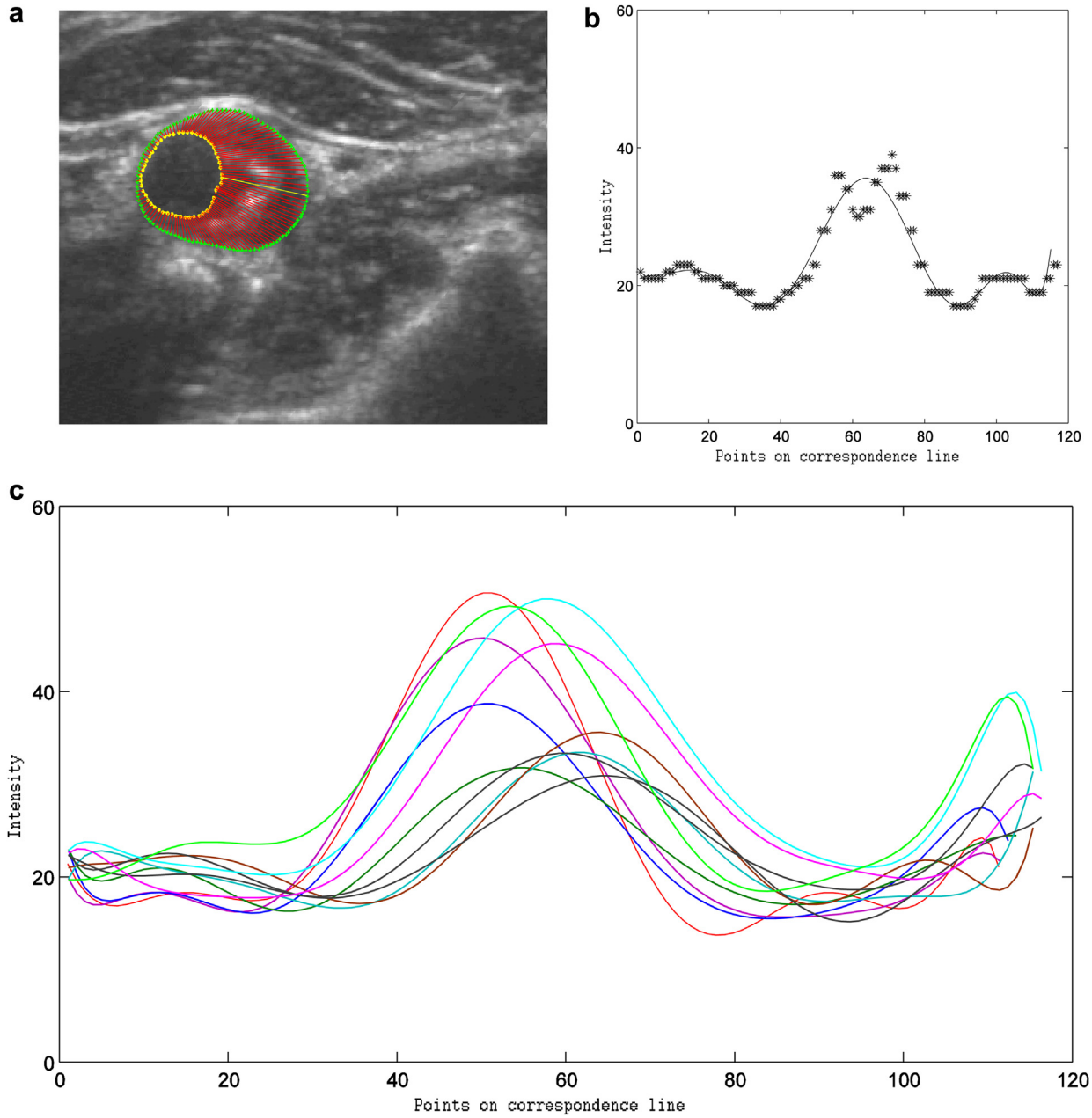

Fig. 3. Example of gray level (GL) values on correspondence lines: (a) carotid image with point correspondence result; (b) GL values on the correspondence line (in yellow in [a]) as a function of distance along the line (pixels), and a polynomial fit to the points; (c) the filtered GL profiles for the five adjacent lines clockwise and counter-clockwise from the yellow correspondence line.

line segments that link the pairs were defined as correspondence lines. The initial plaque "outer" contour points were required to be found on each correspondence line using the gray levels on the correspondence lines. The correspondence lines shorter than 0.4 mm were considered to be spanning a normal vessel wall thickness (Touboul 2007), and LIB points were taken as the initial "outer" contour points for these lines. However, for the other correspondence lines, the sequences of gray levels (GLs) of the pixels on the lines were examined to find the initial "outer" contour of the plaque. To remove the high-frequency spikes, a 10th-order polynomial fit was performed (MATLAB function *polyfit*) for each correspondence line to obtain a smooth filtered curve, as observed in Figure 3b.

Step 2: Searching initial point on filtered curve. In Figures 4, 5 and 7 are three typical intensity profiles between LIB and MAB. In cases with no plaque (Fig. 4), the GL profile appears to be monotonously increasing because the GL of the adventitia is higher than that of the media and intima. However, as observed in Figure 5 with

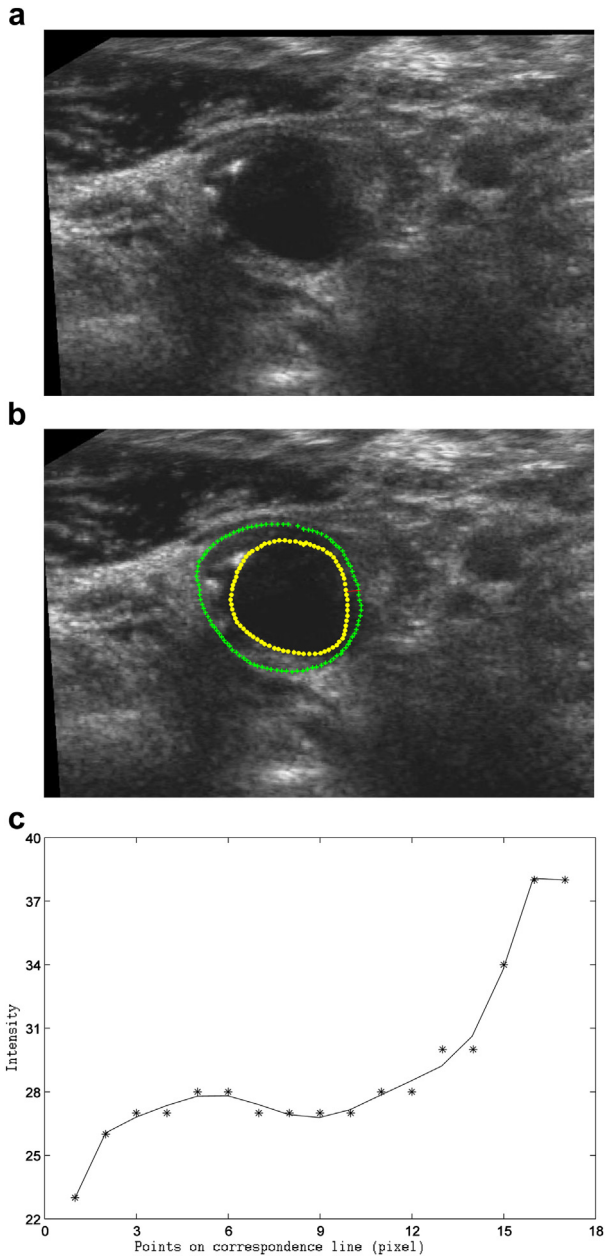

Fig. 4. Example of the intensity profile along the correspondence line in the case with no plaque presence: (a) original image; (b) image with prior knowledge (*i.e.,* media-adventitia boundary in green and lumen-intima boundary in yellow) and a correspondence line in red; (c) gray-level values on the correspondence line (in red in [b]) as a function of distance along the line (pixels), and a polynomial fit to the points.

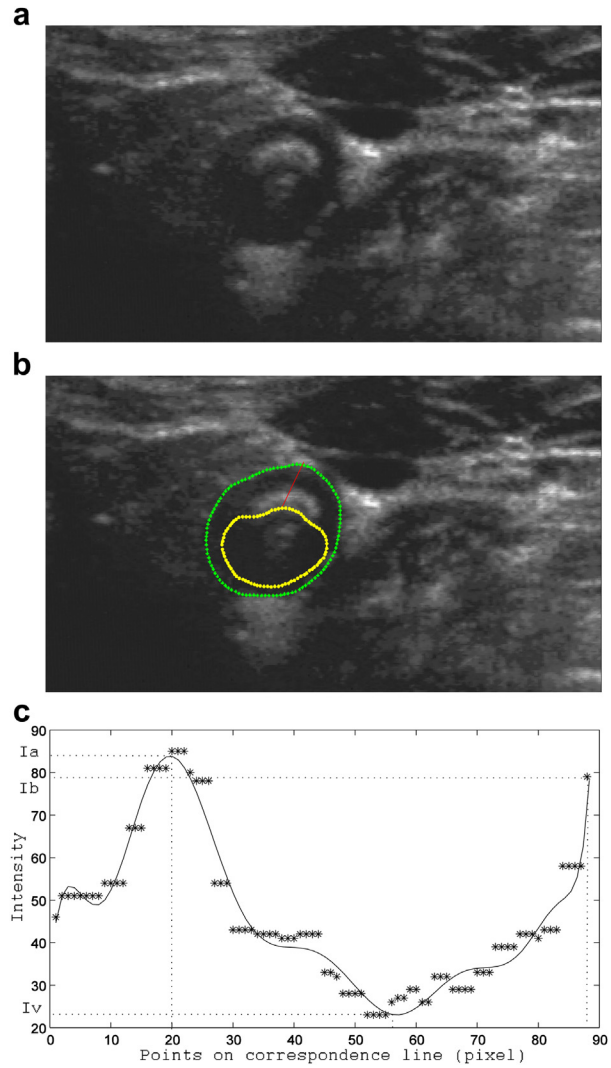

Fig. 5. Example of the intensity profile with presence of both plaque and strong adventitia edge signal: (a) original image; (b) image with prior knowledge (*i.e.,* media-adventitia boundary in green and lumen-intima boundary in yellow) and a correspondence line in red; (c) gray-level values on the correspondence line (in red in [b]) as a function of distance along the line (pixels), and a polynomial fit to the points.

plaque present, the GL profile is sometimes valley shaped, with a minimum (labeled I_v). In this case, every valley point is taken as a candidate for initial plaque "outer" points. The procedure for detection of these initial points consisted of discarding candidate points that do not match the selection criterion (discussed below) and selecting the point that is nearest the MAB. The selection criterion aims to remove "false" valley points caused by normal intima tissue or artifacts and is defined as follows: As indicated in Figure 5c, a typical valley shape has two maxima at locations a and b, with GLs I_a and I_b. Let I_v be the GL at the valley between the two local maxima. The amplitude of GL difference, here defined as $I_a - I_v$, will tend to be larger in brighter regions. Thus, to compensate for this effect, the normalized amplitude of the GL difference is computed as

$$\Delta I = (I_a - I_v)/(I_v + \varepsilon), \tag{1}$$

where ε is a small positive constant used to avoid divisions by zero, and T is the threshold.

Fig. 6. Example of a region with plaque, but the intensity profile has a weak adventitial edge, that is, low image gradient (see *arrow*).

Because of noise and normal intima tissue, other edges can also have a double-peak intensity profile, but the I_a will be weak; thus, ΔI will be small. Hence, to detect the interface between carotid plaque and surrounding media tissue, we consider the candidate valley points with $\Delta I < T$ as "false" points, and the value of T is determined empirically. However, the valley edge feature is not always present at the adventitia boundaries, as shown in the GL profile in Figure 5c. Figure 6 provides an example of a region with plaque, but the intensity profile has a weak adventitial edge. In this case, the normalized GL difference ΔI for the MAB point is computed, as are any other valley points, using eqn (1), and points with $\Delta I < T$ are considered "false" points.

From the remaining candidate points on each filtered curve, the one nearest MAB was chosen as the initial plaque "outer" point. This process was repeated for all correspondence lines and the initial plaque "outer" points were connected to form a closed curve, which is defined as an initial plaque "outer" contour.

2. Level-set evolution. We used the level-set method introduced by Osher and Sethian (1988) to propagate the initial plaque "outer" contour to the desired boundary. In this method, an implicitly given closed curve is iteratively evolved *via* optimization of an objective function to find the desired boundary. Each image is represented as $I : \Omega \rightarrow R$, where $\Omega \subset R^2$ defines the image domain. The evolving curve is represented implicitly as a level set of a 2-D scalar function, referred to as the level-set function $\Phi(x) : \Omega \rightarrow R$, and $x \in \Omega$ is a pixel within the image domain. The boundary is specified as the set of x with $\Phi(x) = 0$. During the segmentation process, the function $\Phi(x)$ is evolved. The level-set evolution process can be expressed by

$$\frac{\partial \Phi(x)}{\partial t} + F|\nabla \Phi(x)| = 0 \qquad (2)$$

where F is the speed function. In our implementation, $\Phi(x)$ is initially defined as a signed distance function of the initial plaque "outer" contour obtained in step 1. Because of the irregular shapes and heterogeneity of

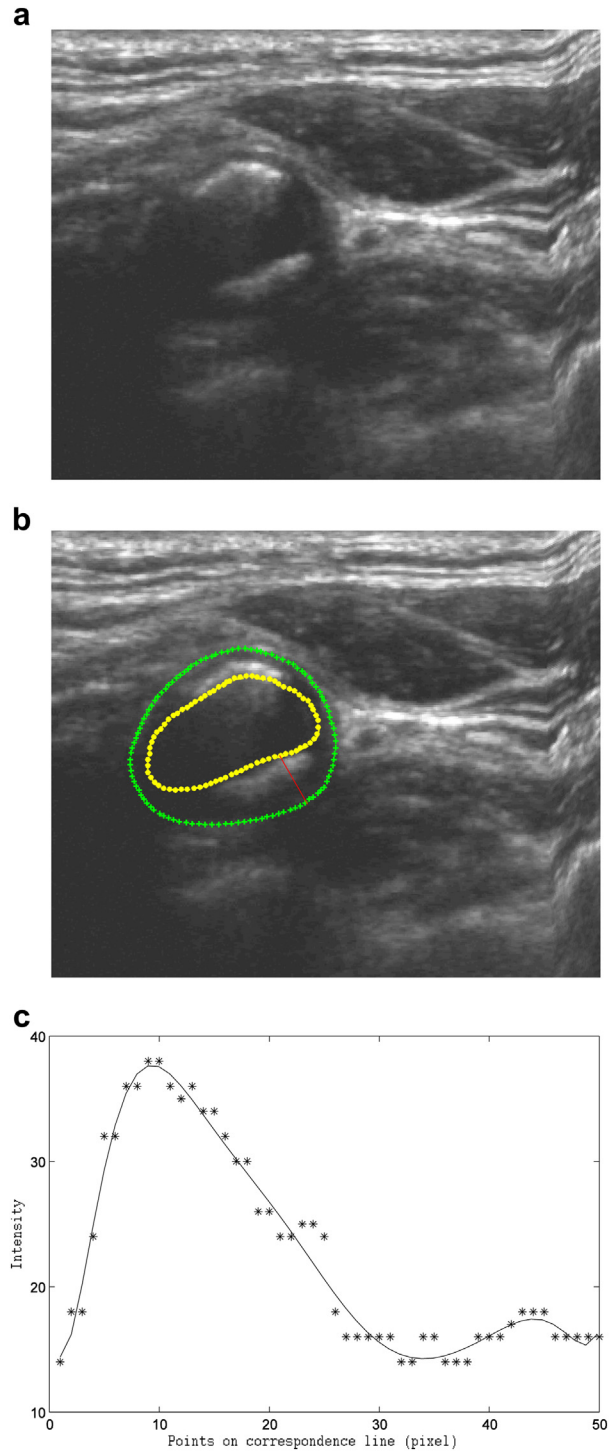

Fig. 7. Example of the intensity profile with plaque presence but weak adventitia edge: (a) original image; (b) image with prior knowledge (*i.e.,* media-adventitia boundary in green and lumen-intima boundary in yellow) and a correspondence line in red; (c) gray-level values on the correspondence line (in red in [b]) as a function of distance along the line (pixels), and a polynomial fit to the points.

plaques, the objective function consists of different energy terms, which incorporate both image GL and anatomic information.

In our work, the sparse field method proposed by Whitaker (1988) was used to efficiently implement the level-set active contours. Updating the level-set evolution only for the neighborhood points of the zero level set, rather than the whole image domain, reduces the computation cost significantly.

The objective function incorporates smoothness energy, a local region-based energy, a global region-based energy and high-level prior knowledge about the possible position of the plaque. The use of these energy terms is based on the consideration that: (i) in general, the interface between the plaque and the media or adventitia has low contrast; (ii) the normal vessel wall has a relatively low GL; and (iii) no significant difference in thickness can be found between the normal vessel wall at different directions in a transverse carotid image. The objective function is defined as

$$E = \lambda_s E_s + \lambda_l E_l + \lambda_b E_b + \lambda_p E_p, \tag{3}$$

Where E_s, E_l, E_b and E_p are the smoothness energy, local region-based Chan-Vese energy (Lankton and Tannenbaum 2008), global Bhattacharyya energy (Michailovich et al. 2007) and position prior-based energy, respectively. λ_s, λ_l, λ_b and λ_p are their weights, which are used to control how severe the driving force provided by the energy terms are. These values are obtained by the optimization process described below (Optimization of Segmentation Parameters section) and are listed in Table 1.

E_s is used to keep the curve smooth by penalizing the arc length of the evolving curve and is given as

$$E_s = \int_\Omega |\nabla H(\Phi(x))| dx. \tag{4}$$

The interior region of the contour is specified by the approximation of the smoothed Heaviside function (Chan and Vese 2001):

$$H(\Phi(x)) = \begin{cases} 1 & \Phi(x) < -\varepsilon \\ 0 & \Phi(x) > \varepsilon \\ \frac{1}{2}\left[1 + \frac{\Phi}{\varepsilon} + \frac{1}{\pi}\sin\left(\frac{\Phi}{\varepsilon}\right)\right] & \text{otherwise} \end{cases}, \tag{5}$$

where ε is a small constant. Similarly, the exterior region of the contour is defined as$(1 - H(\Phi(x)))$.

Because the gradient at the desired boundary is weak, a local region-based energy, instead of an edge-based energy, is incorporated into the objective function:

$$E_l = \int_\Omega \delta(\phi(x)) \int_\Omega B_L(x, \hat{x}) \left[H(\phi(\hat{x}))(I(\hat{x}) - u_x)^2 + (1 - H(\phi(\hat{x})))(I(\hat{x}) - v_x)^2\right] d\hat{x} dx \tag{6}$$

where

$$\delta(\Phi(x)) = \begin{cases} 1 & \Phi = 0 \\ 0 & |\Phi| > \varepsilon \\ \frac{1}{2\varepsilon}\left[1 + \cos\left(\frac{\pi x}{\varepsilon}\right)\right] & |\Phi| \le \varepsilon \end{cases}$$

is the regularized Dirac function. $B_L(x, \hat{x})$ masks a disk-shaped local region with radius r. u_x and v_x represent the GL in the interior and exterior of the contour localized by m$B_L(x, \hat{x})$ at a point x.

The use of the Bhattacharyya energy proposed by Michailovich et al. (2007) is based on the observation that normal vessel tissue is relatively homogeneous and has a low GL. The difference between the two intensity probability density functions (PDFs) inside and outside the contour is increased by the force generated by this energy term. The energy term is given as

$$E_b = B(\Phi(x)) = \int_z \sqrt{P_-(z|\Phi(x)) P_+(z|\Phi(x))} dz, \tag{7}$$

where P_- and P_+ are two smoothed PDFs computed from the global interior and exterior regions of the contour using z intensity bins. The Bhattacharyya coefficient, B, is a measure representing the similarity of the two PDFs. The Bhattacharyya energy aims to differentiate plaques from normal tissue and is computed using GL information of the area between MAB and LIB instead of the whole image domain.

In addition, prior knowledge is also considered to differentiate atherosclerotic plaques from normal vessel tissue. The maximum shrinking distance introduced by Gloger et al. (2009) serves as an estimate of normal vessel thickness and is used as the stopping criterion for the plaque contour evolution in their work. This method was based on the assumption that there exists at least one position at the inner border (*i.e.,* LIB) that does not have a plaque. The closest Euclidean distance from such a location to the outer border (MAB) served as an estimation of normal media thickness. All tissue located between LIB and MAB that had a higher Euclidean distance to the MAB than the normal media thickness delineates a region where plaque is most likely. In Gloger et al. (2009), the normal media thickness was used as a limitation for the evolving curve inside the media and was called maximum shrinking distance (MSD). Experiments indicated that this distance could help to avoid the leakage of the evolving curve into plaque regions. However, a normal vessel without plaque has very small alterations in

thickness. Thus, we incorporated MSD knowledge into the level-set energy function to overcome the thickness alteration issue. For each slice, the MSD was computed as the nearest Euclidean distance between MAB and LIB. The energy term of MSD information is given by

$$E_{\mathrm{p}} = \int_{\Omega} \delta(\Phi(x))P(x)dx \tag{8}$$

where

$$P(x) = \begin{cases} 0 & \|x, \mathrm{MAB}\| \leq \mathrm{MSD} \\ 1 & otherwise \end{cases},$$

$\|x, \mathrm{MAB}\|$ represents the Euclidean distance between x and the MAB. This energy term provides the force to keep the distance between the evolving contour and the MAB within a reasonable range.

To obtain the evolution equation for Φ, we take the derivative of E with respect to Φ. The stopping criterion is reached when the area of the plaque bounded by the evolving contour changes by less than 0.4 mm^2, or a maximum of 1000 iterations is reached. The region between level set-generated contour and LIB denotes the plaque area.

Optimization of segmentation parameters

The optimal values of the segmentation parameters $(\lambda_{\mathrm{s}},\ \lambda_{\mathrm{l}},\ \lambda_{\mathrm{b}},\ \lambda_{\mathrm{p}}$ and $\mathrm{r})$ were determined using the evolutionary strategy (ES) technique (Beyer and Schwefel 2002) and are listed in Table 1. The data set used for optimization consisted of 52 transverse images, 2 images from each subject, which were not used in the validation procedure. We defined the sum of segmentation errors of all optimization images as the objective function to be optimized. In our implementation, the segmentation error for each image is given by the sum of errors (1 – DSC, where DSC = dice similarity coefficient) between the algorithm segmented contour and the mean contour from each expert.

In each ES generation step, 50 sets of segmentation parameters were generated from the set produced by the previous ES generation step by mutation (Beyer and Schwefel 2002). From these, we selected the set of parameters corresponding to the minimum objective function. The sequence of generations was continued until a termination criterion was met. In this work, the energy weights λ_s, λ_{l}, λ_b and λ_{p} were fixed to be within the range 0 to 10; radius r was within the range 4 to 20 pixels. The optimal values of these parameters are listed in Table 1.

Table 1. Optimal values of the algorithm parameters

Description	Parameter	Value
Local region radius	r	4 pixels
Smoothness weight, E_{s}	λ_{s}	8.9
Local region weight, E_{l}	λ_{l}	1.0
Bhattacharyya weight, E_{b}	λ_{b}	1.6
Prior knowledge weight, E_{p}	λ_{p}	4.7

In our study, multi-threaded parallel programming technique was used to reduce the computation time, which still was about 1 d on a PC.

Validation

Validation of the segmentation technique was carried out by comparing the algorithm segmented plaque boundaries with two manual segmentations as described before (Reference Plaque Boundaries section). The validation procedure involved evaluation of 379 transverse slices from 26 common carotid vessels.

Specifically, three types of metrics were used in the evaluation: volume-based metrics, region-based metrics and boundary distance-based metrics. Volume-based metrics represent the segmentation agreement of TPV with manual results. Region-based and boundary distance-based metrics quantify the spatial accuracy of the algorithm segmentations by comparison to manually generated segmentations. Typically, TPV is calculated as the sum of all plaque volumes within a single subject including both the left and right carotid arteries. However, in our evaluation, each plaque was evaluated independently for comparison of manual to algorithm segmentations. Thus, the TPV was computed by multiplying the cross-sectional areas with the inter-slice distance of 1 mm.

Two volume-based metrics were used: TPV difference $(\Delta\mathrm{TPV})$ and absolute TPV difference $(|\Delta\mathrm{TPV}|)$.

$$\Delta\mathrm{TPV}^i = \frac{\mathrm{TPV}_A^i - \mathrm{TPV}_M^i}{\mathrm{TPV}_M^i} \tag{9}$$

$$|\Delta\mathrm{TPV}|^i = \frac{\left|\mathrm{TPV}_{\mathrm{A}}^i - \mathrm{TPV}_{\mathrm{M}}^i\right|}{\mathrm{TPV}_{\mathrm{M}}^i} \tag{10}$$

where $i = 1$ to N_p, where N_p is 34 in this article. $\mathrm{TPV}_{\mathrm{M}}^i$ is the mean measurement of TPVs computed from five repeated measurements for each clinician, and $\mathrm{TPV}_{\mathrm{A}}^i$ is the measurement of TPVs computed using the algorithm. The sign of $\Delta\mathrm{TPV}$ indicates whether the plaque volume is overestimated (+) or underestimated (–).

The mean TPV difference $(\overline{\Delta\mathrm{TPV}})$ and mean absolute TPV difference $(|\overline{\Delta\mathrm{TPV}}|)$ of our data set describe the overall TPV disagreement between the algorithm and manual segmentations:

$$\overline{\Delta\mathrm{TPV}} = \frac{\sum_{i=1}^{N_{\mathrm{p}}} \Delta\mathrm{TPV}^i}{N_{\mathrm{p}}} \tag{11}$$

$$\overline{|\Delta \mathrm{TPV}|} = \frac{\sum_{i=1}^{N_p} |\Delta \mathrm{TPV}|^i}{N_p} \tag{12}$$

The coefficient of variation (COV) is used to compare the variability of manual segmentations for the generation of TPV:

$$\mathrm{COV} = \frac{\overline{\mathrm{SD}}}{\overline{\mathrm{TPV}}} \times 100\% \tag{13}$$

Here

$$\overline{\mathrm{TPV}} = \frac{\sum_{i=1}^{N_p} \mathrm{TPV}^i}{N_p}$$

$$\overline{\mathrm{SD}} = \sqrt{\frac{\sum_{i=1}^{N_p} (\mathrm{SD}^i)^2}{N_p}}$$

SD^i was computed using the five repeated measurements of TPV.

A one-way analysis of variance (ANOVA) was carried out for the TPV differences between the algorithm and manual results, multiple pairwise comparisons were performed using Tukey tests. The ANOVA study was performed with SPSS Version 17.0 (SPSS, Inc., Chicago, IL).

Dice similarity coefficient, plaque area difference (ΔPA) and absolute plaque area difference ($|\Delta \mathrm{PA}|$) were used as region-based accuracy metrics to compare algorithm and manual segmentations. TPA is generally measured in longitudinal carotid images, but for the evaluation here, plaque area was defined as the area enclosed by the individual plaque boundary in 2-D transverse images. The equation of ΔPA is given by

$$\Delta \mathrm{PA}^i = \frac{\mathrm{PA}_A^i - \mathrm{PA}_M^i}{\mathrm{PA}_M^i} \tag{14}$$

where the sign indicates whether the area is overestimated (+) or underestimated (–). We also calculated $|\Delta \mathrm{PA}|$ as the absolute ΔPA value. Similar to volume-based metrics, the mean plaque area difference ($\overline{\Delta \mathrm{PA}}$) and mean absolute plaque area difference ($\overline{|\Delta \mathrm{PA}|}$) were also computed.

The DSC quantifies the area overlap of the algorithm and manual segmentations and is given by

$$\mathrm{DSC} = 2 \times \frac{R_M \cap R_A}{R_M + R_A} \tag{15}$$

where R_M and R_A represent the plaque areas generated by the manual and algorithm methods, respectively. A mean DSC was computed for each pair of manual and algorithm segmentations to provide an overall estimate of area agreement.

We used the mean absolute distance (MAD), maximum absolute distance (MAXD) and average distance (AVD) as boundary distance-based metrics. To compute these metrics, a pointwise correspondence was established by the modified symmetric correspondence algorithm (Chiu et al. 2008a, 2008b). Then, the manual plaque contour, C_M, was expressed as a set of vertices $\{m_i{:}I = 1, 2, \ldots, n\}$, and the set of corresponding vertices on the algorithm-generated contour, C_A, was defined as $\{a_i{:}I = 1, 2, \ldots, n\}$. MAD, MAXD and AVD are given as

$$\mathrm{MAD} = \frac{1}{n} \sum_{i=1}^{n} |d(m_i, a_i)|, \tag{16}$$

$$\mathrm{MAXD} = \max_{i \in [1,n]} \{|d(m_i, a_i)|\} \tag{17}$$

$$\mathrm{AVD} = \frac{1}{n} \sum_{i=1}^{n} d(m_i, a_i) \tag{18}$$

where $d(m_i, a_i)$ is the distance between vertex m_i and its corresponding vertex a_i on the contour C_A.

RESULTS

Examination of the 379 transverse slices from 26 carotid vessels revealed a total of 34 plaques. Of these transverse slices, 117 had no plaques, and in 6, the plaque boundaries could not be obtained using our algorithm method (discussed under Discussion and Conclusions). The training slices were excluded in the validation. The remaining 285 boundaries were used for comparison between manual and algorithm results using all metrics.

Using the sample size calculation with a confidence interval of 0.05, power of 0.8, desired effect size of 5% of the mean of the plaque volumes, that is, about 7.0 mm^3, and standard deviation of 8.0 mm^3, the value calculated for n is about 10. Thus, the sample size of 34 plaques is greater than the number required to observe a change of 5% in the plaque we measured.

Manual segmentations by the expert radiologists and algorithm results on successive images from three patients are shown in Figure 8. This figure indicates that the algorithm-segmented contours are close to the manual contours. Figure 9 gives an overall view of the algorithm segmentations.

The Pearson product-moment correlation analysis was used to evaluate the agreement between the TPVs generated from manual and algorithm segmentations. In Figure 10, the algorithm-generated TPVs are plotted against the manual generated TPVs of each expert. High correlation coefficients of 0.993 and 0.992 were found for segmentations by experts 1 and 2, respectively. Figure 10 also illustrates the variations (mm^3) in TPVs using Bland-Altman Plots. The mean bias is -5.3 mm^3

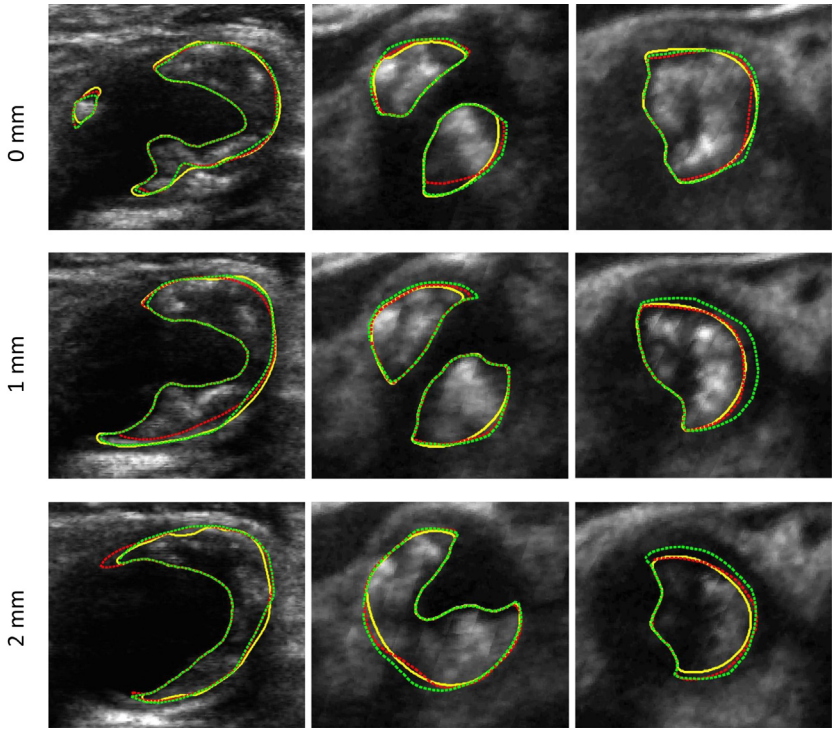

Fig. 8. Sample results of algorithm and manual segmentations for three patients (one subject in each column). The panels in the rows were obtained at distances of 0, 1 and 2 mm from the carotid bifurcation. Yellow solid contours = algorithm results, red dashed contours = results of expert 1, green dashed contours = results of expert 2.

for the TPVs generated by expert 1, and the mean bias for expert 2 is -8.5 mm^3.

The agreement between PAs computed from manual and algorithm segmentations is illustrated in Figure 11. Pearson correlation coefficients for experts 1 and 2, compared with the algorithm results, are 0.983 and 0.980 respectively. For expert 1, the bias is -0.6 mm^2 with a 95% confidence interval (CI) of -4.9 to 3.7 mm^2, and for expert 2, the bias is -1.0 mm^2 with a 95% CI of -5.1 to 3.1 mm^2.

Table 2 summarizes the overall evaluation results including ΔTPV, |ΔTPV|, ΔPA, |ΔPA| DSC, MAD, MAXD and AVD used to evaluate the degree of discrepancy between the area and distance measurements of the algorithm and manual segmentations. This table also lists the difference between the two experts (Expert 1–expert 2 column).

We used the COV values to compare the intra-observer variability in estimating the TPV. The results of the two experts yielded COV values of 5.66% and 7.85%, respectively.

We also performed a one-way ANOVA study of multiple pairwise comparisons with an α value of 0.05, and obtained a p-value of 0.95, indicating that we failed to show a statistically significant difference between TPVs of the algorithm and two experts using 34 individual plaques.

In Figure 12 are graphs of the COVs of five repeated manual segmentations as a function of mean plaque volume, and Figure 13 illustrates the effect of plaque volume on DSC values.

The segmentation times were determined using the 379 slices five times. About 5 to 7 min of operator time was required to manually outline a single plaque once in a 3-D US image. The algorithm required a mean of 4.9 ± 1.3 s to segment a single image using a non-optimized MATLAB implementation on a single core of a PC with processor speed of 1.8 GHz.

DISCUSSION AND CONCLUSIONS

Carotid plaque segmentation from 3-D US images is an important task for monitoring plaque progression and regression. Compared with manual segmentation, the proposed algorithm resulted in segmentation time savings.

A segmentation method based on the level-set framework requires an operator to place several points to initialize the algorithm, which introduces segmentation variability. Our algorithm eliminates the need for operator interaction by using MAB and LIB as prior knowledge. Thus, an operator involved in image processing should be able to reproduce the method. In this work, the position information of the plaque is

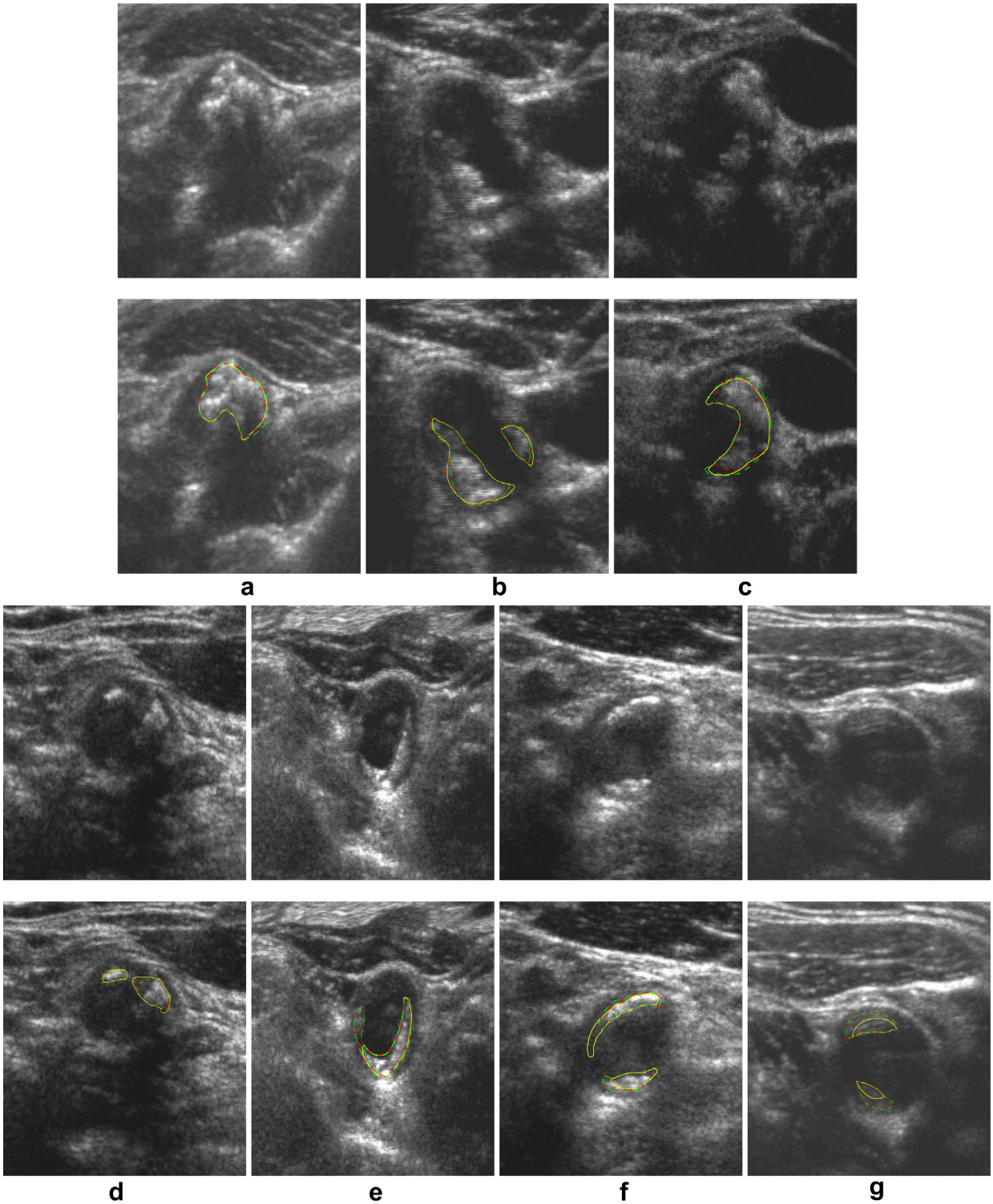

Fig. 9. Original images and the same images with segmented result superimposed. Yellow solid contour = algorithm result, red or green dashed contour = manual result. The images are shown in order of decreasing dice similarity coefficient from (a) to (g).

incorporated with the GL information between MAB and LIB into an initialization step as well as the level-set evolution.

The MSD method introduced by Gloger et al. (2009) was used in our algorithm as an energy term to maintain the evolving contour within a reasonable distance range

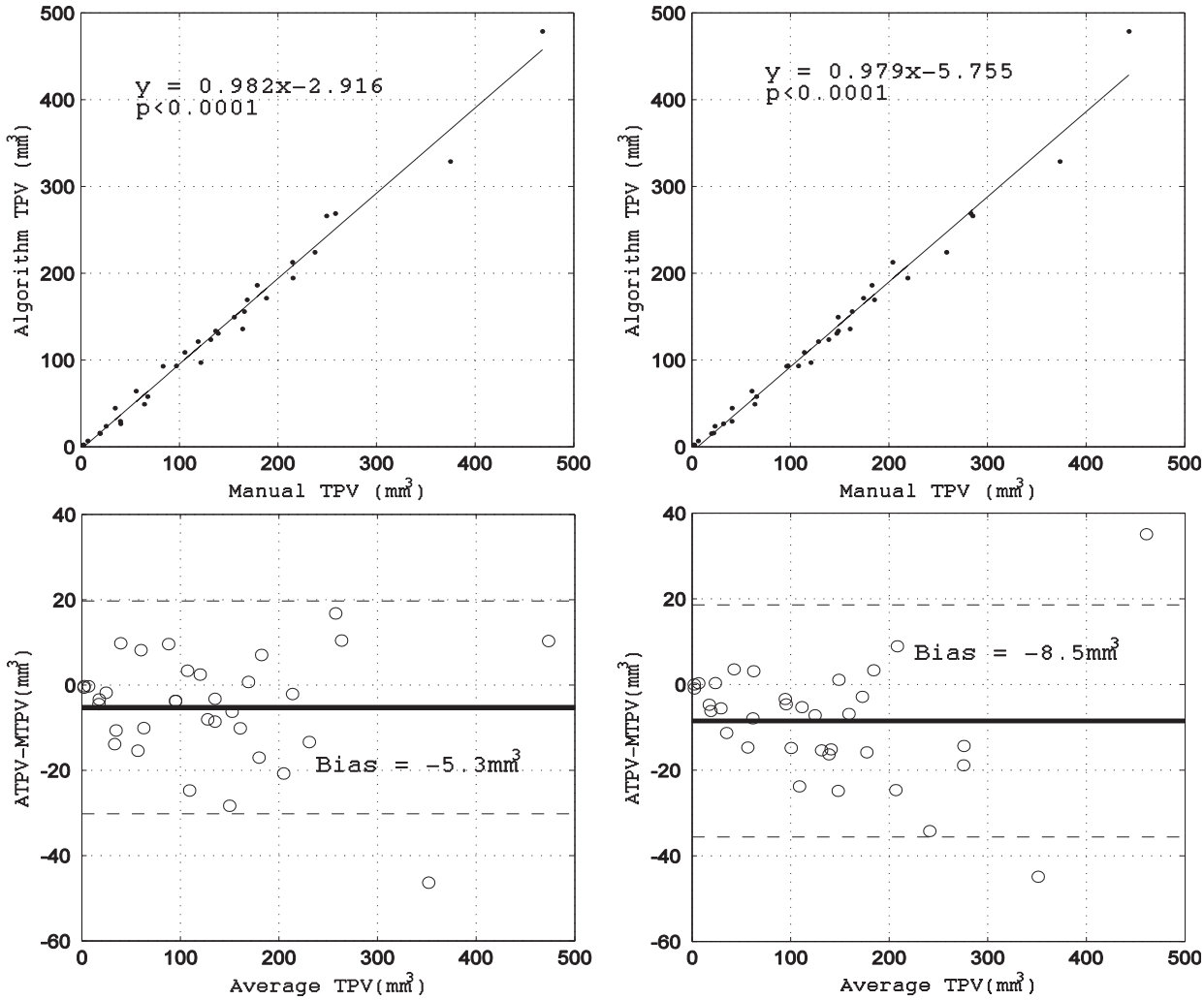

Fig. 10. Correlation and Bland-Altman plots of the total plaque volume (TPV) generated from the proposed method. Column 1 = expert 1, column 2 = expert 2.

from the MAB. This term provides a force to restrict the evolving contour from leaking for plaques with low echogenicity. Another important advantage of this algorithm is that it is suitable for all 2-D slices extracted from a carotid 3-D US image, including the bifurcation slice. Compared with the correspondence method making use of a polar coordinate conversion, the modified point correspondence algorithm can deal with complex shapes and established correspondence between MAB and LIB in all our cases.

The validity of our algorithm was investigated by comparing the algorithm results with manual traced results from two experts. ANOVA failed to show a statistical significant difference with the data set we used. Bland-Altman and Pearson correlation plots indicated that the algorithm slightly underestimates both plaque volume and plaque area compared with manual segmentation. In addition, the algorithm results are closer to the manual result from expert 1 than that from expert 2. The mean coefficients of variation (5.66% and 7.85%) indicate that expert 1 generated more stable and reliable results compared with expert 2.

Segmentation accuracy is quantitatively evaluated in Table 2. Overall DSC values of 88.5% and 87.9%, compared with expert 1 and expert 2, respectively, indicate that the segmentations provided by the algorithm have a large spatial overlap with those provided by the manual approach. The results of all metrics are consistent with the Bland-Altman result that the segmentation has higher agreement with expert 1. The, ΔPA and AVD values indicate that our algorithm obtained results with smaller plaque regions than the manual method. This may be due to the fact that the boundary driven by the level-set energy is liable to stop where there is a significant difference in gray level between the outside and inside of the plaque, while experts tend to consider the

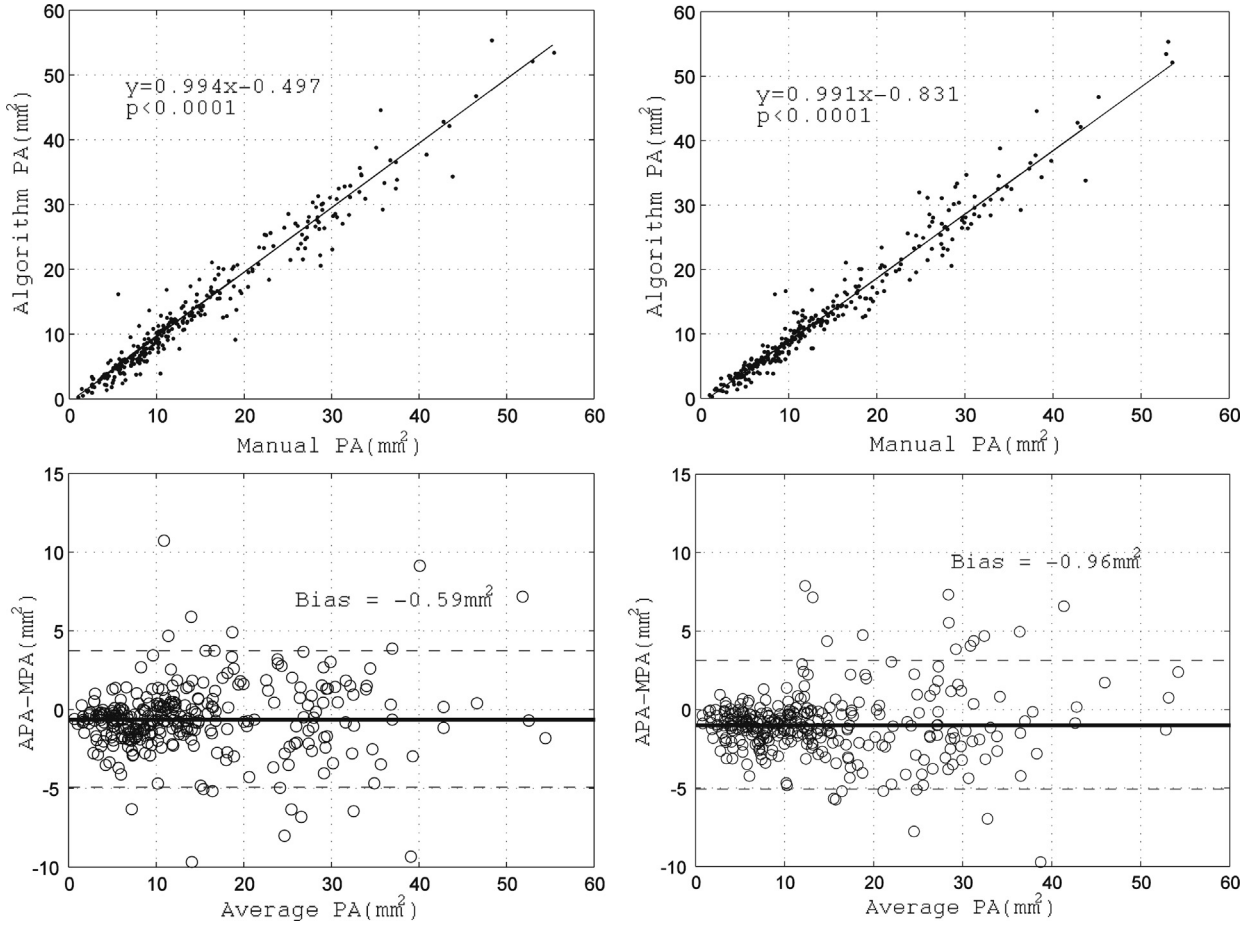

Fig. 11. Correlation and Bland-Altman plots of the plaque area (PA) generated from proposed method. Column 1 = expert 1, column 2 = expert 2.

region where the grey level starts to change as a part of the carotid plaque.

We also examined the intra-observer COV in the measurement of plaque volume as a function of the mean plaque volume for the 34 individual plaques. Ainsworth et al. (2005) reported that as plaque volume increases, the COV in the measurement of plaque volume tends to decrease. Our values obtained for intra-observer variability, which range from about 28% to 2%, are comparable to the corresponding variability reported by Ainsworth et al. (2005). The COV ranged from 21.2% to 3.9% for the smallest to the largest plaque volumes (2.5 to 468.3 mm^3) measured by expert 1. For the range of plaque volumes measured by expert 2 (2.3 to 443.6 mm^3), the COV ranged from 28.1% to 2.4%.

In addition, we examined DSCs as a function of plaque volume. The DSC values exhibited a smaller error compared with manual results as plaque volume increased.

The outlining of a typical plaque, consisting of 10 to 30 slices, takes about 5 to 7 min. A 3-D carotid US image with multiple plaques will take longer. In contrast, time needed for our algorithm to segment a 3-D image is about 3 min, reducing operator time by about 60%. Compared with manual segmentation, the algorithm always underestimated the plaque. In addition, the expert observers identified six 2-D plaque boundaries that the algorithm

Table 2. Summary of all segmentation accuracy metrics comparing algorithm and manually generated boundaries by the experts

Metric	Algorithm-expert 1*	Algorithm-expert 2*	Expert 1-expert 2†
ΔTPV(mm^3)	−5.3 ± 12.7‡	−8.5 ± 13.8	−3.3 ± 11.2
|ΔTPV|(mm^3)	9.9 ± 9.5	11.8 ± 11.1	8.0 ± 8.4
ΔPA(mm^2)	−0.6 ± 2.2	−1.0 ± 2.1	−0.4 ± 2.2
|ΔPA|(mm^2)	1.6 ± 1.6	1.7 ± 1.6	1.4 ± 1.8
DSC (%)	88.5 ± 8.2	87.9 ± 8.3	92.1 ± 8.4
MAD (mm)	0.20 ± 0.19	0.20 ± 0.18	0.17 ± 0.20
MAXD (mm)	0.78 ± 0.88	0.75 ± 0.89	0.64 ± 0.82
AVD (mm)	−0.09 ± 0.21	−0.11 ± 0.19	−0.07 ± 0.21

* Difference between algorithm segmentation and mean contour computed from 5 manual segmentations by expert 1 (expert 2).

† Difference in the metrics between the manually traced contours of the two experts.

‡ Mean ± standard deviation.

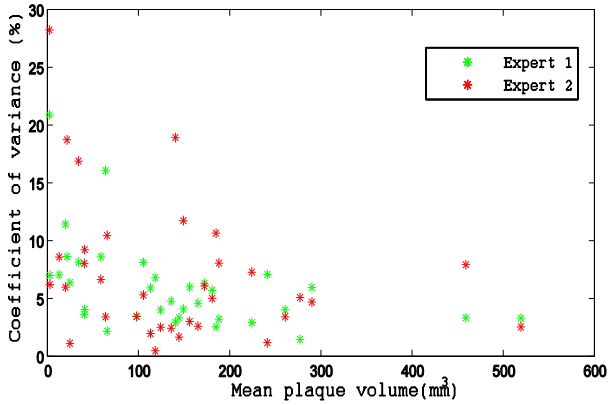

Fig. 12. Coefficients of variance as a function of plaque volumes.

missed. These six contours could not be detected using our method because of the poor contrast of the small plaque cross sections in transverse views. All six plaque contours were located at an end slice of a 3-D plaque. This effect was also found and analyzed by Landry et al. (2007). Because adjacent slices are highly correlated, incorporating information from longitudinal views may overcome this limitation. The number of longitudinal views could be consistent with the number of radial lines between MAB and LIB.

In our method, we sliced the reconstructed 3-D images into 2-D images 1 mm apart. Segmentation of the original 2-D US images may generate better results, as reconstruction and re-slicing degraded image quality. However, use of the original images would be more time consuming because the acquired 2-D images are separated by 0.2 mm. Fenster et al. (2006) studied the effect of inter-slice distance for measuring the carotid plaque volume in 3-D ultrasound images. They found that a 1-mm inter-slice distance was suitable for plaque volume measurement. So segmenting the re-sliced images rather than the original 2-D images was a compromise considering execution time and segmentation accuracy.

The accuracy of our algorithm was investigated by comparing the algorithm results with the manually traced results of two experts. Manual measurement of carotid plaque volume using 3-D ultrasound images was validated as an accurate and reproducible method by Landry et al. (2004). Also, because it is very difficult to reproduce the appearance of plaque in animals or phantoms as found in patients, the variability must be studied in patient data.

Because our plaque segmentation method follows the segmentation of MAB or LIB, we used the segmentations that were available to us, namely, manual MAB and LIB (Landry et al. 2004). As inaccurate segmentations of the MAB or LIB may lead to a segmentation error, our next phase in this work will be to combine a method for automatic segmentation of MAB and LIB and our algorithm in a single software program to realize automatic analysis of carotid atherosclerosis. The flattening map

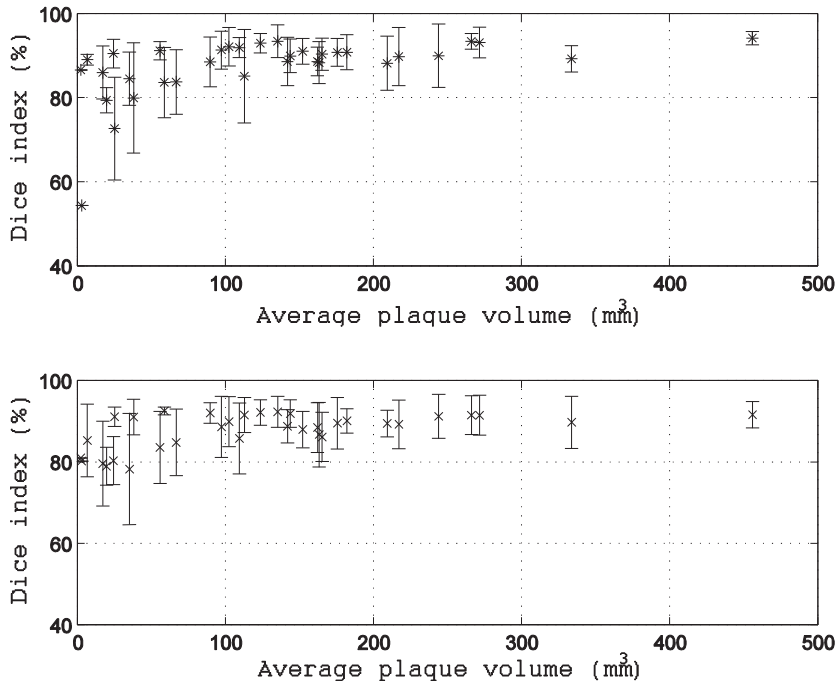

Fig. 13. Dice similarity coefficients as a function of average plaque volumes compared with two manual segmentations. Top graph = expert 1, bottom graph = expert 2.

approach (Chiu et al. 2008a, 2008b; Egger et al. 2008) could then be used to superimpose segmented plaques onto maps of vessel wall plus plaque thickness to monitor spatial details of carotid atherosclerosis progression and regression.

In level-set evolution, the driving forces produced by the four energy terms cooperate to provide the composite force. In cases of ulceration, the MSD energy will restrict the evolving contour from shrinking too much, while the local region energy will produce the opposite force because of the intensity difference between the vessel tissue and plaque. To validate the algorithm on ulcerated plaques, 3-D US images with ulcerations that have been manually identified are needed to test our approach for quantifying and detecting ulcerations. This will be the subject of future studies.

In this study, we only segmented the CCA including the bifurcation. However, in the future, we will investigate the segmentation of internal and external carotid arteries as well. Plaque progresses along the vessel (in the axis of flow) 2.4 $\times$ faster than it thickens and also grows circumferentially (Barnett et al. 1997). So we need to ensure that the region for segmentation is sufficiently large to detect change in all dimensions (Landry et al. 2007) to provide more accurate estimates for monitoring plaque change.

In conclusion, the proposed level-set method combining prior and image information produced segmentation results close to the manual contours of the two experts. Our approach reduces the need for operator interaction and generates measurements with the necessary precision and accuracy for clinical use. Total plaque volumes and other related measurements can be obtained when plaque contours are acquired.

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